

Assessing the Reliability of AI-Driven Motor Competence Systems and Investigating Accuracy Through Expert Informed Refinement

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INTRODUCTION

Motor competence (MC) is an essential component of childhood development and is associated with participation in physical activity, physical fitness and broader health outcomes [1,2]. Despite its importance, MC is low in Australian children [3]. To address low levels, accurate assessment is vital. Numerous valid, and reliable assessments are available, yet they lack ecological validity in real world settings [4]. As such, assessment is rarely conducted and remediation is rare [5]. Artificial intelligence (AI)-driven assessment systems may provide a scalable alternative by automating MC assessment for children. Moving the Next Generation (MNG) is a novel AI-driven assessment that has been developed, with preliminary reliability demonstrating promise [6]. However, the accuracy and reliability of MNG in real world settings has not been determined.

METHODS

Using a 3-stage methodical approach this study aims to assess the accuracy and reliability of MNG and whether its accuracy can be improved through expert-informed refinement. Stage 1 will involve using video-based MC assessment data of children aged 4-12, performing hopping and jumping motor skills. The data will be assessed independently through manual coding and MNG using the criteria determined by TGMD-3 [7], following assessment agreement will be determined between the two assessment types using inter-rater reliability and Cohen's kappa. Stage 2 will correct the incorrect outputs of MNG identified in the agreement. The newly corrected assessments will then be manually integrated into MNG's training data to establish a refined MNG model. Stage 3 will use the refined MNG model to assess newly collected video-based data. Manual coding of the newly collected data will also

occur to determine agreement/performance of the newly refined MNG model.

EXPECTED FINDINGS

It is anticipated that this study will evaluate the reliability of the current MNG platform and highlight areas requiring improvement. Expert-informed refinement is expected to improve the accuracy and consistency of AI-generated outcomes through the development of a refined model. Findings will contribute to the development of more accurate and reliable automated MC assessment with the potential to support more accessible assessment in school and community settings.

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